

LaboSignes: an Interactive French Sign Language Recognition Interface

Jules Françoise
Université Paris-Saclay, CNRS,
Laboratoire Interdisciplinaire des
Sciences du Numérique
Orsay, France
jules.francoise@lisn.fr

Julie Lascar
Université Paris-Saclay, CNRS,
Laboratoire Interdisciplinaire des
Sciences du Numérique
Orsay, France
jlascar@lisn.fr

Cyril Verrecchia
Université Paris-Saclay, CNRS,
Laboratoire Interdisciplinaire des
Sciences du Numérique
Orsay, France
cyril.verrecchia@lisn.fr

Sidonie Minodier
Université Paris-Saclay, CNRS,
Laboratoire Interdisciplinaire des
Sciences du Numérique
Orsay, France
sidonie.minodier@universite-paris-saclay.fr

Michèle Gouïffès
Université Paris-Saclay, CNRS,
Laboratoire Interdisciplinaire des
Sciences du Numérique
Orsay, France
michele.gouiffes@universite-paris-saclay.fr

Annelies Braffort
Université Paris-Saclay, CNRS,
Laboratoire Interdisciplinaire des
Sciences du Numérique
Orsay, France
annelies.braffort@lisn.fr

Abstract

This article presents the design, implementation and evaluation of LaboSignes, a web-based interface for interactive French Sign Language (LSF) recognition, designed to enable search-by-sign for Deaf users. We iteratively designed a gesture-based recording interface, and integrated a lightweight Transformer model enabling low-latency recognition. We evaluated three recording interfaces (gesture-based, posture-based, and button) through a user study with 16 LSF signers. Participants preferred the gesture-based interface, despite comparable recognition accuracy across variants. The study also brings insights into sources of errors, with participants emphasizing the need to accommodate regional sign variations to support linguistic diversity in SL technology design.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Accessibility systems and tools**.

Keywords

Sign Language, Recognition, Interface Design, Empirical Study.

ACM Reference Format:

Jules Françoise, Julie Lascar, Cyril Verrecchia, Sidonie Minodier, Michèle Gouïffès, and Annelies Braffort. 2026. LaboSignes: an Interactive French Sign Language Recognition Interface. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799328>



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3799328>

1 Introduction

Access to education and information remains particularly difficult for Deaf¹ individuals due to language barriers. Despite the official recognition of Sign Languages (SLs) – the natural languages of Deaf communities, – most online content, such as lexical resources, news websites, and forums, remains inaccessible through sign-based search [12]. As a result, users are compelled to rely on their local written language to navigate information.² Despite progress in Artificial Intelligence (AI) for written and spoken language, interactive systems enabling SL-based interaction remain scarce. First, automatic SL processing is challenging because of the complexity of SLs' grammar involving multiple articulators (hands, body and face), and of data scarcity [3]. Second, few studies explore the design and evaluation of user interfaces for sign-language recognition from an HCI standpoint [17]. The few sign-recognition systems available online [7, 11, 22] are documented from a technical viewpoint but often lack user evaluations.

In this paper, we describe the design of a web-based interactive sign recognition system for French Sign Language (LSF) that works from any webcam. The resulting prototype, called *LaboSignes*, integrates a custom sign recording interface that provides postural feedback and uses gesture recognition to enable hands-free recording. The system was optimized for efficiency, with real-time keypoint extraction and fast classification using a lightweight transformer model. Our interface facilitates results exploration and provides interactive 3D motion capture recordings for the signs. To evaluate this interface, we conducted a study with 16 LSF signers to compare the usability of various recording interfaces, and assess the performance of a sign classifier. Results show that the gesture-based interface was preferred, despite no significant difference in recognition accuracy, and that participants value regional variations in sign articulation.

¹The capital D represents the membership of the linguistic and cultural community, not the auditory status.

²This issue is particularly problematic for many Deaf individuals whose primary language is a sign language (SL), as they may face difficulties with reading and writing in a written language acquired as a second language.

2 Related Works

2.1 Background on Automatic Sign Languages Processing

Automatic SL processing is particularly challenging because of SL unique spatio-temporal and illustrative grammar,³ and the continuous and highly multidimensional nature of the data [3]. Developing SL processing tools therefore requires advancing Machine Learning (ML) algorithms for automatic sign recognition (SLR) [2, 4], spotting [14], or translation between SLs and written languages [3]. These tasks are particularly challenging due to the linguistic complexity of SLs and limited data availability. In this paper, we consider the task of Isolated Sign Language Recognition (ISLR), that aims to identify and classify individual signs performed by a signer into a corresponding written word or label. SLR is necessary to enable applications involving search by sign. Previous works on ISLR use either RGB-based methods – which achieve high performance scores but rely on heavier models [4] – and pose-based approaches that use a skeleton representation of the signer, resulting in lighter models suited for interactive recognition [2, 13]. These advances in automatic SL processing are typically assessed using the few available datasets, leaving research on designing and evaluating such tools largely underexplored within HCI [17].

2.2 User Studies Informing or Evaluating SLR Systems

In recent years, researchers have highlighted persistent power imbalances in SL technology, where Deaf individuals are rarely directly involved [5, 8]. This oversight creates biases, such as treating SLs as if they could be captured by gloss sequences like spoken languages, or building tools that fail to meet the actual needs of the intended users. Recently, HCI studies have been conducted to inform the design of SL technologies from the perspectives of Deaf users regarding, for instance, their motivations and expectations for SL technologies [21, 23].

Yet, despite developments in automatic SL recognition and translation, few interactive systems enabling search by sign are publicly available, and user evaluations of recognition systems are lacking. For example, only 3 platforms are publicly available for sign recognition: for Peruvian SL [11], Belgian SL [7], or healthcare in Spanish SL [22]. Few works have conducted user studies to evaluate sign recognition, in the context of healthcare scenarios [19], or with a restricted vocabulary with inexperienced signers [18]. Other studies with human participants use Wizard-of-Oz approaches to compare touch and SL input [20], to assess the impact on user experience of classifier accuracy [1], or to explore hybrid search involving linguistic properties to complement imperfect recognition [10].

In this paper, we describe the iterative design of an interactive ISLR interface for French Sign Language (LSF), and evaluate the performance of the recognition system through a user study with deaf and hearing signers.

³Signers use a three-dimensional signing space in front of them to position concepts, objects, and actors, allowing these entities to interact and be referenced later. Signers employ both conventional (lexical) signs – which are standardized enough to appear in dictionaries but can vary with the context and signer – and illustrative (depicting) signs, which are created on the spot to convey movement, size, location, or relationships.

3 Designing a Sign Recognition Interface

Our design process involved a multidisciplinary team composed of hearing researchers in HCI and SL processing, and a Deaf engineer specialized in SL animation. After training the sign recognizer, we adopted an iterative design process for the web interface, testing each version with the entire team to refine our design.

3.1 Design Goals

The long-term goal of this research is to advance sign-based search in online resources, including bilingual dictionaries, keyword-based search in media content, and specialized terminological resources – particularly for scientific lexicons. With this project, we focused on designing and evaluating an interactive recognition system from the perspective of potential users. This raised several design goals for the recognition interface:

- (1) To ensure broad accessibility, the system should avoid relying on specialized equipment like depth cameras or data gloves. Instead, it should use only conventional webcams, and leverage standard web technologies compatible with a wide range of devices and operating systems.
- (2) The recognition system should be robust and adaptable to a wide range of users and usage contexts. This includes accommodating diverse demographics, backgrounds, lighting conditions, and variations in user positioning.
- (3) The interface should be designed to enable Deaf participants to query by sign, incorporating an accessible recording mechanism, ensuring low-latency recognition to maintain interactivity, and providing a clear visualization interface to help users explore alternative results effectively.

3.2 Sign Recognition Model

We started by collecting and annotating a dataset of LSF signs for recognition. We extracted lexical signs from the Mediapipe-RGB dataset [16] using the method described in [14]. The resulting dataset contains approximately 15,000 video clips of signs across 445 classes spanning diverse categories (locations, numbers, sports, politics). The signs were extracted from continuous signing sequences recorded under studio conditions, sourced from an online French Sign Language (LSF) media platform. Consequently, the vocabulary primarily pertains to news topics.

As our design goals emphasize robustness to variations in visual domain and low latency, we decided to use a pose-based approach with a lightweight Transformer model for the recognition. Each video is preprocessed using Mediapipe holistic landmark detection [15]. Then, a subset of keypoints of body, hands and face are selected and rescaled. Sequences are normalized in length using truncation or padding. Finally, the classification is performed by a Transformer encoder with 2 layers and 4 attention heads. The full technical description of the recognition method is available in Lascar et al. [13].

3.3 Designing the Sign Recording Interface

We designed and implemented a web interface for ISLR using the model described above. However, the initial recording interface lacked visual feedback on the user's posture – a critical limitation, as the quality of keypoint extraction relies on the visibility of all

joints, particularly the torso, hands, and face, within the frame. Additionally, the interface required users to interact with buttons to start and stop recordings, which disrupted their natural posture. We iterated on the design of the recording interface, focusing on two key dimensions: **posture feedback** and **recording triggering mechanisms**. Existing systems often use visual cues such as circles [7] or silhouettes [11] to guide user positioning within the video frame, while recording is typically initiated via a button with a countdown or automatically when the user's face and hands are detected within designated areas [7].

In our exploration, we experimented with visual feedback mechanisms designed to assess the user's posture based on their distance from the camera and the presence of specific keypoints (e.g., wrists) with high confidence. However, we found that complex criteria relying on Mediapipe's joint visibility scores could introduce instability and degrade the user experience, particularly when strict posture control interfered with the ability to record. To address this, we adopted a simpler approach for posture feedback: a gray disk that turns green when the user's head is positioned within it and their distance from the camera falls within an optimal range. This ensures that most of the typical signing space remains visible.

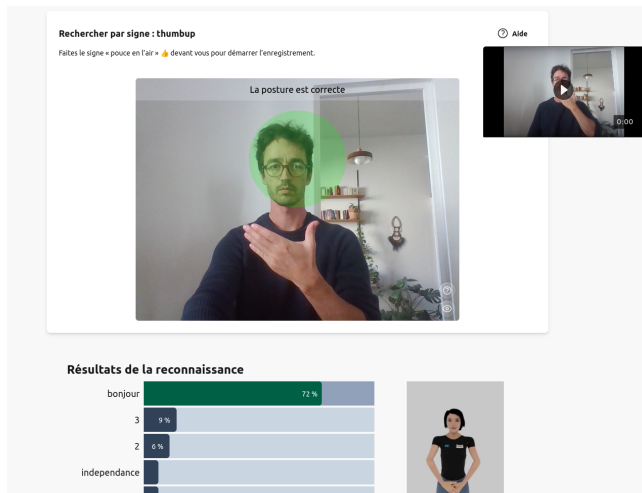


Figure 1: Screenshot of the system's interface. The recording interface (top) includes a disk that turns green when the user is at the correct distance and body position, together with a message at the top. Recordings are triggered by a thumb-up gesture. At the bottom, recognition results are displayed using an interactive bar chart. Clicking on a result plays a reference recording of the sign.

For recording, we tested both posture-based triggering and gesture recognition. Ultimately, we opted for gesture-based triggering, as it provided users with control, avoiding involuntary triggering. In our final design, the recording begins after a 3-second countdown initiated by the recognition of a "thumb-up" gesture. The recording automatically stops when both hands exit the frame or when the total pixel motion drops below a predefined threshold. The final interface is shown in Figure 1 (top).

3.4 Visualizing Results and Reference Signs

The model outputs a probability distribution across all 445 known classes. In our system, the five most probable classes are visualized as an interactive bar chart (see Figure 1, bottom). Users can click on any bar to play a reference video of the corresponding sign, allowing them to compare the model's top predictions alongside their associated probabilities.

For a subset of 196 signs, we recorded full-body motion capture data of the third author performing each sign, to provide interactive 3D avatars. The process is outlined in Figure 2. We used an Optitrack marker-based optical motion capture system composed of 24 cameras (Flex 3 and 13) connected to the Motive software. The signer wore a full Optitrack's motion capture suit with 41 passive retro-reflective markers, as well as Manus gloves. After cleaning and processing the recordings, the data was rendered as an avatar. We designed four different avatars for the platform. The web interface provides a detailed view for each sign, featuring an interactive 3D viewer of the animated avatar (Figure 2 (d)). This tool serves as a pedagogical aid, enabling learners to visualize and explore signs in a dynamic 3D space.

3.5 Implementation

Our interface is a web application implemented using the Marcelle toolkit [9]. The client is a single-page application built with SvelteKit, that communicates over WebSocket with a Node.js server for data storage, and over HTTP with a Python server for inference. To optimize the reactivity of the recognition, we ported the keypoint extraction, selection and normalization to JavaScript. Mediapipe's web implementation allows for real-time client-side keypoint extraction, meaning that the sequence of selected keypoints can be readily available at the end of the recording, reducing latency. On a standard PC (Intel i5 2.40GHz CPU; 16GB of RAM), inference latency (including JavaScript keypoint preprocessing and inference on a Python server) is about 21 ± 7 ms. In practice, network latency often has a bigger impact than inference time. The source code of the recording interface and prototype is available online.⁴

4 User Study

We conducted a within-subject study to evaluate the performance of our sign recognition system across three variants of the recording interfaces. We focused on assessing overall accuracy and understanding preferences, challenges and opportunities with the recording interface.

4.1 Methods

4.1.1 Participants. We recruited 16 participants (10 women, 6 men), including 10 Deaf individuals and 6 interpreters or experienced signers. One participant was left-handed. Recruitment was conducted through the researchers' professional networks and during a workshop focused on Sign Language for mathematics education organized by the Sign'Maths⁵ group. Participation was voluntary, and no financial compensation was offered. All participants provided written informed consent, which was also translated into French Sign Language (LSF) by professional interpreters.

⁴<https://gitlab.lisn.upsaclay.fr/mtals/publications/chi2026-poster-labosignes>

⁵<https://signmaths.univ-tlse3.fr/>

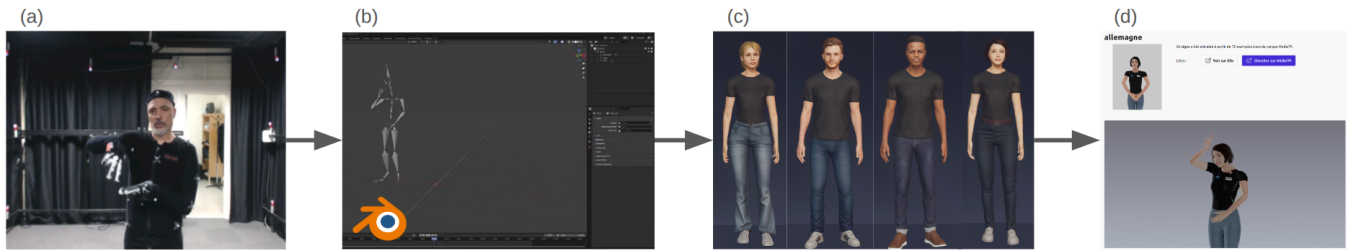


Figure 2: Overview of the avatar creation process. Each sign is recorded using motion capture (a). Recordings are cleaned and refined (b), then rendered on avatars (c). The 3D models are integrated in the web platform with an interactive viewer (d).

4.1.2 Setup and Procedure. Our study employed a within-subject design, with the recording interface as the primary factor, comprising three conditions. We adapted the interface described in Section 3.3 to create three variants, each using the same 3-second countdown and automatic stopping, but differing in how the recording was triggered:

thumb-up : triggering once a "thumb-up" gesture is recognized.

auto : triggering once the user is in the correct posture, after activating the webcam with a button, as in [7].

button : triggering using a simple button.

Participants were asked to record 10 signs with each interface. The list of 30 signs⁶ was selected from the 445-sign vocabulary to reflect the diversity of signs (one or two-handed, with or without repetitions, with or without a highly distinctive form in relation to other signs). Interface order was counterbalanced across participants. Signs were elicited using a word in French and a visual from the IVT visual dictionary⁷ depicting the sign. Then, participants could experiment freely with the system. We gathered qualitative feedback through written comments and informal discussions following the study with the assistance of an interpreter if needed.

4.1.3 Data Collection and Analysis. 25 videos were excluded due to recording issues, leaving 455 instances across 30 labels. To evaluate the system's performance, we calculated two accuracy metrics: top-1 and top-5 per-instance accuracies.⁸ We analyzed performance across gender, hand dominance, and auditory status (deaf vs. hearing participants). After verifying that our data did not meet the assumptions of normality using Shapiro-Wilk tests, we employed Mann-Whitney U tests for non-parametric comparisons. In the post-study questionnaire, we gathered data on users' preferred interface, interfaces with which the recording was inadvertently triggered, and interfaces that participants found particularly difficult to activate. Additionally, we collected open-ended feedback and documented observations from post-study discussions with participants.

⁶The order of signs was randomized once, in three blocks. **Block 1**: 'bonjour', 'chien', 'village', 'euro', 'fini', 'enfant', 'photo', 'pluie', 'ministre', 'semaine'. **Block 2**: 'libre', '2', 'septembre', 'premier', 'malade', 'ville', 'femme', 'juin', 'president', 'recherche'. **Block 3**: 'magasin', 'masque', 'directeur', 'sourd', 'noel', 'monde', 'journée', 'place', 'soir', 'france'.

⁷<https://ivt.fr/editions/dictionnaires>

⁸During the study, we observed significantly lower performance for the left-handed participant, which revealed a bug in the data augmentation pipeline—specifically, in generating symmetric versions of each video. Following this observation, we retrained the model using the same procedure, and the updated results are presented in this paper.

4.2 Results

4.2.1 Recording Interface Preferences. Our study revealed distinct preferences among participants for the three recording interfaces tested. The **thumb-up** interface emerged as the most favored, with 10 participants preferring it, compared to 3 for the **auto** interface and 2 for the **button** interface. However, usability challenges were noted: 7 participants reported involuntary triggering with the auto interface, while 3 encountered the same issue with the thumb-up interface. Additionally, 4 participants struggled to intentionally trigger recording using the thumb-up interface, highlighting potential areas for design refinement. Following the free exploration phase, participants suggested that the posture guide could be improved by increasing the required distance, noting that the current configuration sometimes resulted in a restricted signing space. Participants reported that the gesture-based interface was advantageous during the free exploration phase, as it enabled them to seamlessly record multiple signs in a row without interruption.

4.2.2 Recognition Accuracy. The offline evaluation of the recognizer on the MediaPi dataset (445 labels) resulted in a top-1 accuracy of 90.87%, rising to 98.56% for top-5 accuracy, demonstrating robust performance. On the user study's data (30 test labels out of 445), performance reached a top-1 accuracy of 71.65% and a top-5 accuracy of 87.03%. These results mean that the correct label appeared in the visualization interface more than 87% of time, underlining the model's robustness to domain shift – from coarticulated signs recorded in a professional environment, to isolated signs recorded with a webcam. Using Mann-Whitney U-tests, we did not find any significant difference in accuracy according to the recording interface variant, gender or hand dominance. However, we observed a difference in accuracy according to the auditory status, with higher accuracy for Deaf participants. A Mann-Whitney U-test revealed a significant difference in top-1 accuracy between the two groups ($U = 10.00, p = 0.031, z = -2.12$, moderate effect size $r = 0.53$).

4.2.3 Sources of Errors. To understand the potential sources of errors of the recognition model, we visualized all misclassified videos and analyzed qualitative feedback. While not all errors are interpretable, we identified several key sources: (1) the presence of another person in the video, or of parasite movements at the end of the recording, (2) the failure to consider depth in some gestures (e.g. with forward or backward movement expressing the future or the past for week days), (3) linguistic ambiguities (e.g. the signs for "dog" and "Bordeaux" are almost identical in LSF), and

(4) local/regional variants of the sign articulation. This regional variation is particularly pronounced in France, where signs were developed in various educational institutions across the country. Delaporte [6] argues that this diversity stems from a process of dialectization: 19th-century Parisian signs, initially disseminated by educators, diverged regionally due to the disruptive impact of the Milan Congress (1880), which promoted oralism and marginalized sign-based instruction. Most participants actually emphasized the need to account for regional variations to preserve the linguistic richness of their language.

5 Discussion and Future Directions

We presented the design of an interactive sign recognition system, and its evaluation with both deaf and hearing LSF signers. Our results show that lightweight pose-based transformers are robust to changes in visual domain, making them suitable for webcam-based search. Overall, participants preferred our interface enabling gesture-based recording, despite similar recognition accuracy. Participants' reflections helped clarify the causes of errors, such as nuances in movement dynamics and regional variations. These variations introduce complex linguistic, political, and ethical challenges, with participants highlighting the need to prevent automatic SL processing from exacerbating language standardization at the expense of diversity. In future research, designing participation interfaces could help expand the diversity of SL data and preserve the language's richness, while also empowering the Deaf community to take an active role in shaping AI technologies for SL processing.

Acknowledgments

This work was supported by the French National Research Agency (ANR) through the GestToSay project (ANR-25-CE23-4683). It was performed using HPC resources from GENCI-IDRIS (Grant 2024-AD011015441). We thank Sign'Maths for allowing us to organize a workshop, as well as participants for their time and valuable feedback. We also thank Médiapi for sharing bilingual resources.

References

- [1] Oliver Alonzo, Abraham Glasser, and Matt Huenerfauth. 2019. Effect of Automatic Sign Recognition Performance on the Usability of Video-Based Search Interfaces for Sign Language Dictionaries. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility*. ACM, Pittsburgh PA USA, 56–67. doi:10.1145/3308561.3353791
- [2] Matyáš Boháček and Marek Hruš. 2022. Sign Pose-Based Transformer for Word-Level Sign Language Recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*. 182–191.
- [3] Danielle Bragg, Oscar Koller, Mary Bellard, Larwan Berke, Patrick Boudreault, Annelies Braffort, Naomi Caselli, Matt Huenerfauth, Hernisa Kacorri, Tessa Verhoef, Christian Vogler, and Meredith Ringel Morris. 2019. Sign Language Recognition, Generation, and Translation: An Interdisciplinary Perspective. In *The 21st International ACM SIGACCESS Conference on Computers and Accessibility*. ACM, Pittsburgh PA USA, 16–31. doi:10.1145/3308561.3353774
- [4] Joao Carreira and Andrew Zisserman. 2017. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [5] Maartje De Meulder. 2021. Is “Good Enough” Good Enough? Ethical and Responsible Development of Sign Language Technologies. In *Proceedings of the 1st International Workshop on Automatic Translation for Signed and Spoken Languages (AT4SSL)*. Association for Machine Translation in the Americas, Virtual, 12–22.
- [6] Yves Delaporte. 2005. La Variation Régionale En Langue Des Signes Française. *Marges Linguistiques* 10 (“Langues régionales”, présenté par Claudine Moise, Véronique Fillol, Thierry Bulot) (2005), 118–132.
- [7] Jerome Fink, Pierre Poitier, Maxime André, Loup Meurice, Benoît Frénay, Anthony Cleve, Bruno Dumas, and Laurence Meurant. 2023. Sign Language-to-Text Dictionary with Lightweight Transformer Models.. In *IJCAI*. 5968–5976.
- [8] Neil Fox, Bencie Woll, and Kearsy Cormier. 2023. Best Practices for Sign Language Technology Research. *Universal Access in the Information Society* (2023). doi:10.1007/s10209-023-01039-1
- [9] Jules Françoise, Baptiste Caramiaux, and Téo Sanchez. 2021. Marcelle: Composing Interactive Machine Learning Workflows and Interfaces. In *The 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21)*. Association for Computing Machinery, New York, NY, USA, 39–53. doi:10.1145/3472749.3474734
- [10] Saad Hassan, Akhter Al Amin, Alexis Gordon, Sooyeon Lee, and Matt Huenerfauth. 2022. Design and Evaluation of Hybrid Search for American Sign Language to English Dictionaries: Making the Most of Imperfect Sign Recognition. In *CHI Conference on Human Factors in Computing Systems*. ACM, New Orleans LA USA, 1–13. doi:10.1145/3491102.3501986
- [11] Joe Huamani-Malca, Miguel Rodríguez Mondoñedo, Francisco Cerna-Herrera, Gissella Bejarano, Carlos Vásquez Roque, Cesar Augusto Ramos Cantu, and Sabina Oporto Pérez. 2024. Lessons from Deploying the First Bilingual Peruvian Sign Language - Spanish Online Dictionary. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue (Eds.). ELRA and ICCL, Torino, Italia, 10316–10323. <https://aclanthology.org/2024.lrec-main.901/>
- [12] Amélie Josselin-Leray and Jérémie Segouat. 2024. *Les dictionnaires de langue des signes française à l'ère du numérique: un premier état des lieux*. Éditions de l'Université de Sherbrooke, 117–148.
- [13] Julie Lascar, Jules Françoise, Michèle Gouiffès, Annelies Braffort, and Diandra Fabre. 2026. PoET: Lightweight Pose Encoder Transformer for Online Sign Language Recognition. In *Proceedings of the 21st International Conference on Computer Vision Theory and Applications (VISAPP'26)*, Vol. 1). 19–28.
- [14] Julie Lascar, Michèle Gouiffès, Annelies Braffort, and Claire Danet. 2024. Annotation of LSF subtitled videos without a pre-existing dictionary. In *LREC-COLING 2024 11th Workshop on the Representation and Processing of Sign Languages: Evaluation of Sign Language Resources*. 100–108.
- [15] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Ubweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, Wan-Teh Chang, Wei Hua, Manfred Georg, and Matthias Grundmann. 2019. MediaPipe: A Framework for Building Perception Pipelines. (2019). arXiv:1906.08172 [cs.DC] <https://arxiv.org/abs/1906.08172>
- [16] Yanis Ouakrim, Hannah Bull, Michèle Gouiffès, Denis Beauteemps, Thomas Hueber, and Annelies Braffort. 2024. Mediapi-RGB: Enabling Technological Breakthroughs in French Sign Language (LSF) Research Through an Extensive Video-Text Corpus.. In *Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications*. SCITEPRESS - Science and Technology Publications, Rome, Italy, 139–148. doi:10.5220/0012372600003660
- [17] Soraia Prieth, J. Alfredo Sánchez, and Josefina Guerrero. 2022. A Systematic Review of User Studies as a Basis for the Design of Systems for Automatic Sign Language Processing. *ACM Transactions on Accessible Computing* 15, 4 (Dec. 2022), 1–33. doi:10.1145/3563395
- [18] Cristian A. Restituyo, Fernando Loizides, Shufan Yang, Kurtis Weir, Adam Worrallo, Thomas Hartley, Nicos Souleles, and Michael Loizou. 2017. Presenting and Investigating the Efficacy of an Educational Interactive Mobile Application for British Sign Language Using Hand Gesture Detection Techniques. In *Proceedings of the 16th World Conference on Mobile and Contextual Learning*. ACM, Larnaca Cyprus, 1–8. doi:10.1145/3136907.3136945
- [19] Muhammed Süzgün, Hilal Özdemir, Necati Camgöz, Ahmet Kindiroğlu, Doğaç Başaran, Cengiz Togay, and Lale Akarun. 2015. Hospisign: an interactive sign language platform for hearing impaired. *Journal of Naval Sciences and Engineering* 11, 3 (2015), 75–92.
- [20] Nina Tran, Paige S DeVries, Matthew Seita, Raja Kushalnagar, Abraham Glasser, and Christian Vogler. 2024. Assessment of Sign Language-Based versus Touch-Based Input for Deaf Users Interacting with Intelligent Personal Assistants. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 53, 15 pages. doi:10.1145/3613904.3642094
- [21] Nina Tran, Richard E. Ladner, and Danielle Bragg. 2023. U.S. Deaf Community Perspectives on Automatic Sign Language Translation. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*. ACM, New York NY USA, 1–7. doi:10.1145/3597638.3614507
- [22] Manuel Vázquez-Enríquez, José Luis Alba-Castro, Ania Pérez-Pérez, Carmen Cabeza-Pereiro, and Laura Docío-Fernández. 2024. SignaMed: a Cooperative Bilingual LSE-Spanish Dictionary in the Healthcare Domain. In *Proceedings of the LREC-COLING 2024 11th Workshop on the Representation and Processing of Sign Languages: Evaluation of Sign Language Resources*. 386–394.
- [23] Adele Vogel and Jessica L Korte. 2024. What Factors Motivate Culturally Deaf People to Want Assistive Technologies?. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–7. doi:10.1145/3613905.3650871